EAST WEST UNIVERSITY
Department of Computer Science and Engineering

CSE 345 Project Report

Course Name: Digital Logic Design
Course Code: CSE 345
Section No: 2

Lab Project

Project Title : Design and Implementation of 2 Bit Full Adder Circuit using Multiplexer

Submitted To:

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Introduction:

A full adder is a combinational logic circuit used to perform binary addition. In this project, a 2-bit full adder is implemented using the cascade method, where two 1-bit full adders are connected in series. The carry output from the first stage is used as the input carry for the second stage, eliminating the need for an external carry input.

The circuit takes two 2-bit binary numbers (A1 A0) and (B1 B0) as inputs and produces two sum outputs (Sum1, Sum2) along with a final carry output.

The design is implemented using multiplexers and basic logic gates:

- An 8:1 multiplexer is used for sum1
- 2:1 multiplexers are used for generating sum2
- And and Or gates for Carry generation (C_{final})

This approach demonstrates how complex arithmetic circuits can be constructed using multiplexers and cascading techniques, reducing circuit complexity and improving design efficiency.

A_1	A_0	B_1	B_0	C	Sum_2	Sum_1
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

Table 1: Truth Table 2 Bit Full Adder

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

Table 2: Truth Table for Multiplexer (8:1)

For an 8:1 multiplexer with select lines S_2, S_1, S_0, each input $I_k$is selected based on the binary value of $(S_2S_1S_0)$. The correct Boolean expression uses minterms of the select lines.

Here's the properly written equation:

$$Y = \bar{S_2}\,\bar{S_1}\,\bar{S_0}\,I_0 + \bar{S_2}\,\bar{S_1}\,S_0\,I_1 + \bar{S_2}\,S_1\,\bar{S_0}\,I_2 + \bar{S_2}\,S_1\,S_0\,I_3 + S_2\,\bar{S_1}\,\bar{S_0}\,I_4 + S_2\,\bar{S_1}\,S_0\,I_5 + S_2\,S_1\,\bar{S_0}\,I_6 + S_2\,S_1\,S_0\,I_7$$

Multiplexer:

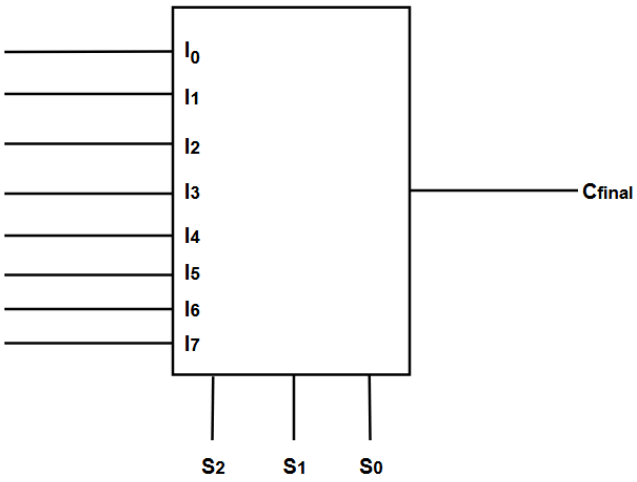


Figure 1: Multiplexer for Carry

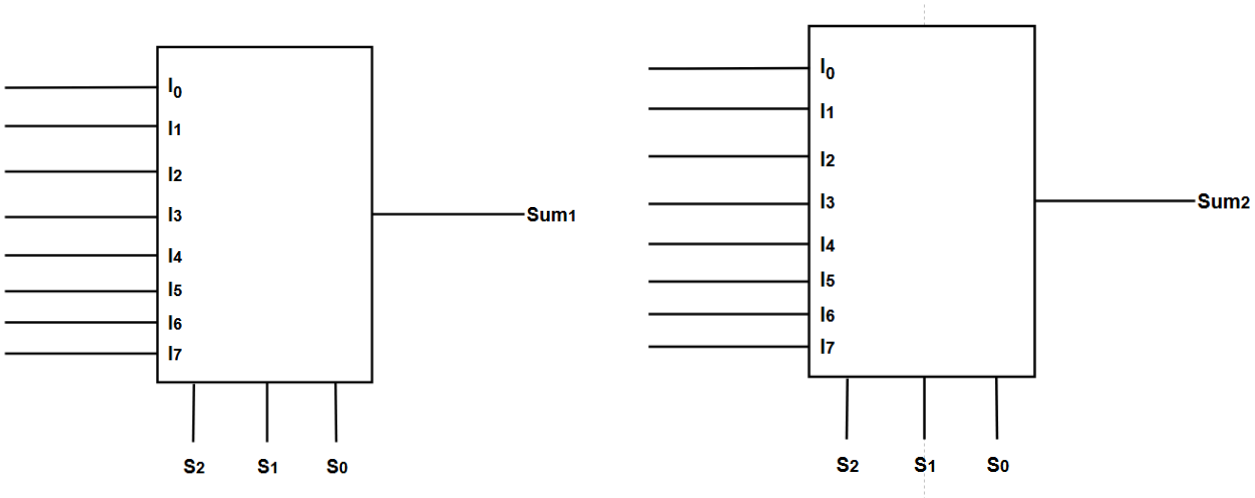


Figure 2: Multiplexers for Sum_1 and Sum_2

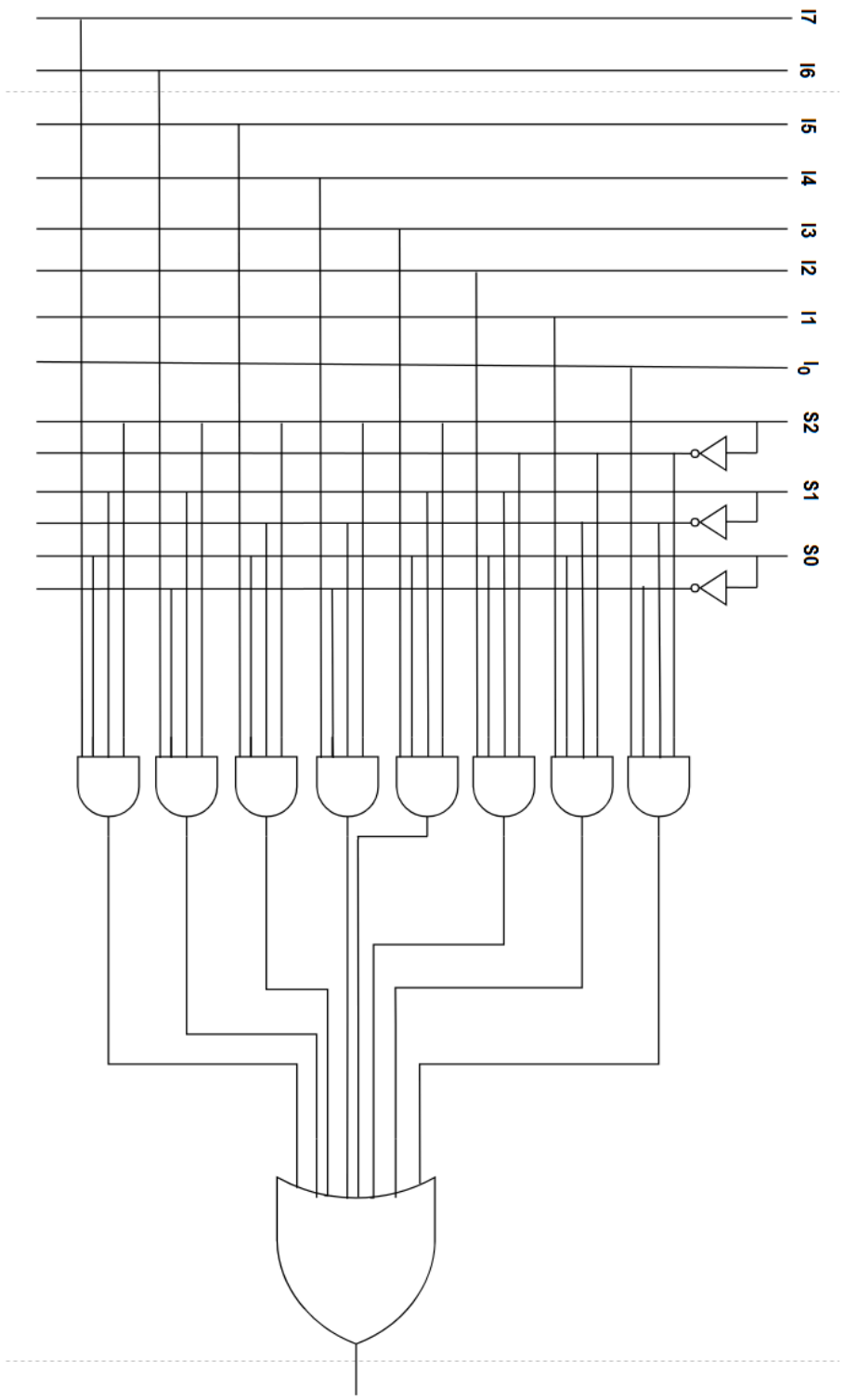

I7
I6
I5
I4
I3
I2
I1
I0
S2
S1
S0

Figure 3: Logical Circuit Diagram

A_1	A_0	B_1	B_0	C_{final}	
0	0	0	0	0	0
0	0	0	1	0	
0	0	1	0	0	0
0	0	1	1	0	
0	1	0	0	0	0
0	1	0	1	0	
0	1	1	0	0	B_0
0	1	1	1	1	
1	0	0	0	0	0
1	0	0	1	0	
1	0	1	0	1	1
1	0	1	1	1	
1	1	0	0	0	B_0
1	1	0	1	1	
1	1	1	0	1	1
1	1	1	1	1	

Table 3: Truth Table for Carry

A_1	A_0	B_1	B_0	Sum_2	
0	0	0	0	0	0
0	0	0	1	0	
0	0	1	0	1	1
0	0	1	1	1	
0	1	0	0	0	B_0
0	1	0	1	1	
0	1	1	0	1	$\underline{B}_0$
0	1	1	1	0	
1	0	0	0	1	1
1	0	0	1	1	
1	0	1	0	0	0
1	0	1	1	0	
1	1	0	0	1	$\underline{B}_0$
1	1	0	1	0	
1	1	1	0	0	B_0
1	1	1	1	1	

Table 4: Truth Table for Sum_1

A_1	A_0	B_1	B_0	Sum_1	
0	0	0	0	0	B_0
0	0	0	1	1	
0	0	1	0	0	B_0
0	0	1	1	1	
0	1	0	0	1	$\underline{B}_0$
0	1	0	1	0	
0	1	1	0	1	$\underline{B}_0$
0	1	1	1	0	
1	0	0	0	0	B_0
1	0	0	1	1	
1	0	1	0	0	B_0
1	0	1	1	1	
1	1	0	0	1	$\underline{B}_0$
1	1	0	1	0	
1	1	1	0	1	$\underline{B}_0$
1	1	1	1	0	

Table 5: Truth Table for Sum_2

Equipment and Components Needed:

1. 8 to 1 mux ic × 1
2. 2 to 1 mux ic × 3
3. 4 input and ic × 2
4. 4 input or ic × 1
5. 2 input or ic × 1
6. Breadboard
7. Wires
8. Trainer Board

Working Principle:

The 2-bit full adder is implemented using the cascade method, where addition is performed in two stages from LSB to MSB. The circuit is realized using multiplexers instead of conventional logic gates.

In the first stage, the least significant bits (A0) and (B0) are added. Since there is no external carry input, the initial carry is assumed to be 0. The outputs of this stage are:

- $Sum1 = A0 + B0$
- Carry (C1 = A0 +B0) when both is A0 = 1 and B0 = 1 [it is calculated internally]

These outputs are generated using **8:1 multiplexers**, where the select lines control the selection of appropriate input combinations based on the truth table.

In the second stage, the most significant bits (A1) and (B1) are added along with the carry (C1). The outputs of this stage are:

- $Sum2 = A1 + B1 + C_1$ [Previously calculated C1]
- Final Carry ($C_{final} = A + B + C_1$) [Final Carry is Generated when A1 =1 and B1=1]

The **2:1 multiplexer** is used to calculate Sum2 generate the carry output. The input variables act as select lines, and the data inputs (I0) to (I7) are assigned according to the truth table. Based on the select line combination, the correct output is selected, implementing the carry function.

Thus, by properly configuring the multiplexers, the required sum and carry outputs are obtained. The cascade structure ensures that the carry generated from the first stage is correctly propagated to the second stage, completing the 2-bit addition.

Schematic Diagram:

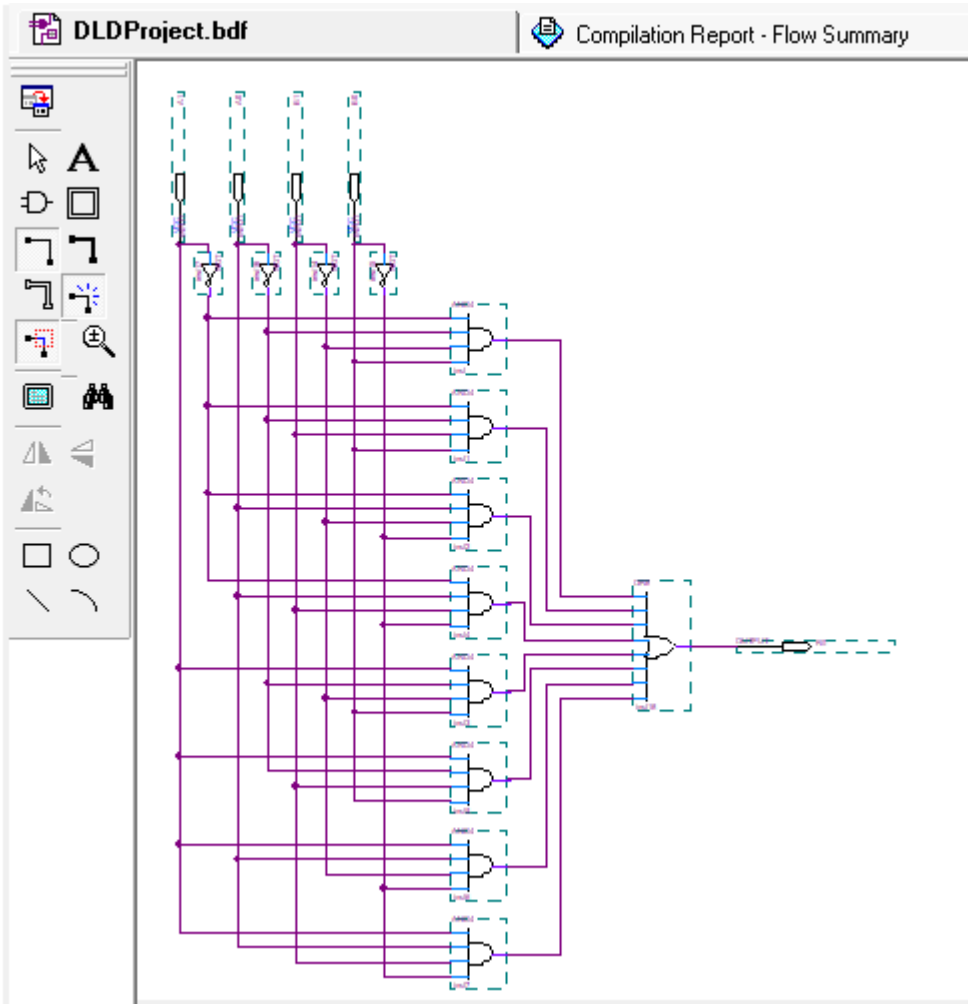


Figure 7: Logical Circuit Diagram (Simulation)

Schematic Simulation:

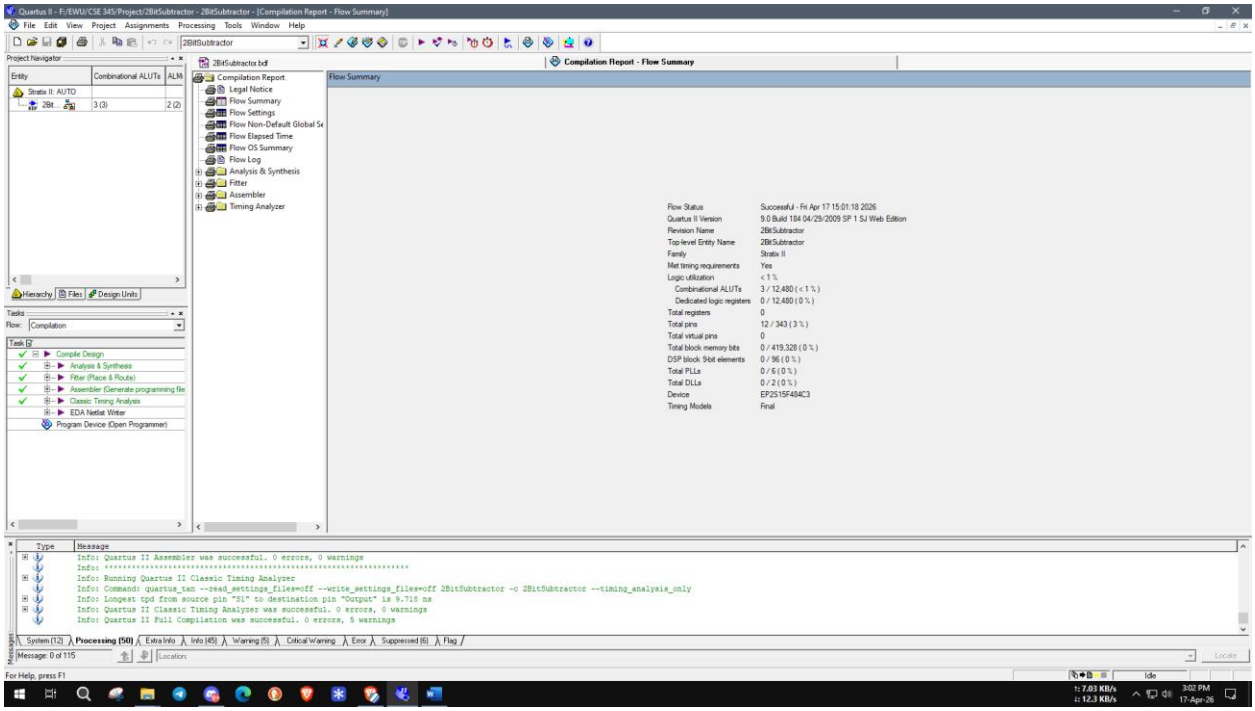


Figure 8: Compilation Report

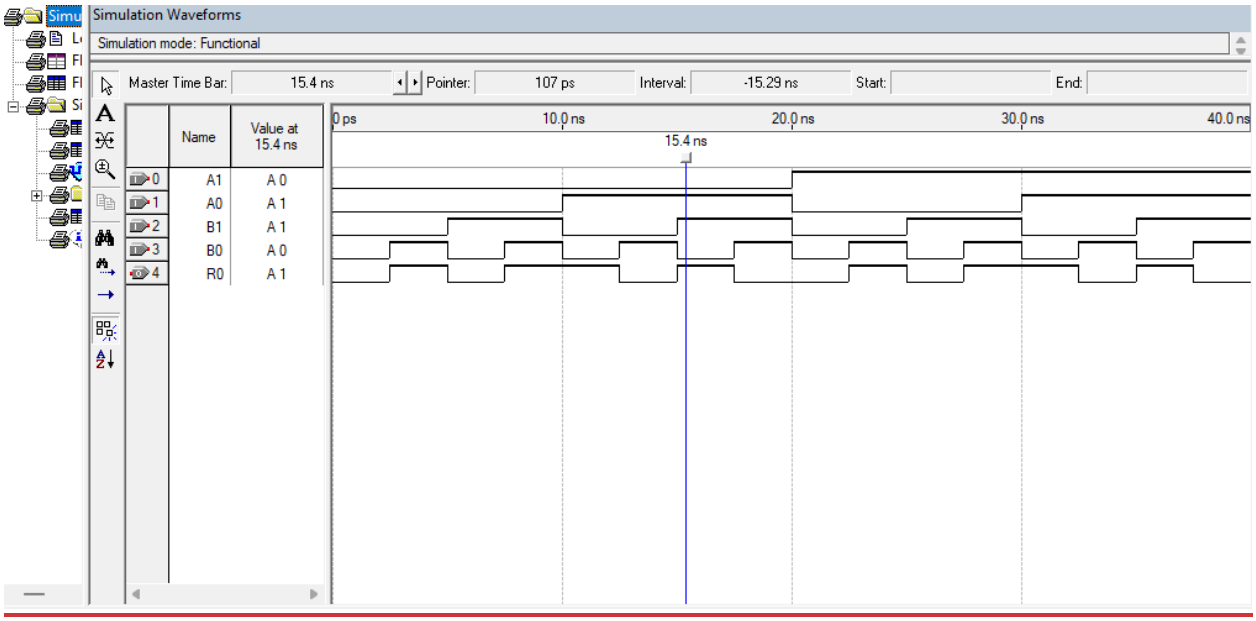


Figure 9: Input waveform

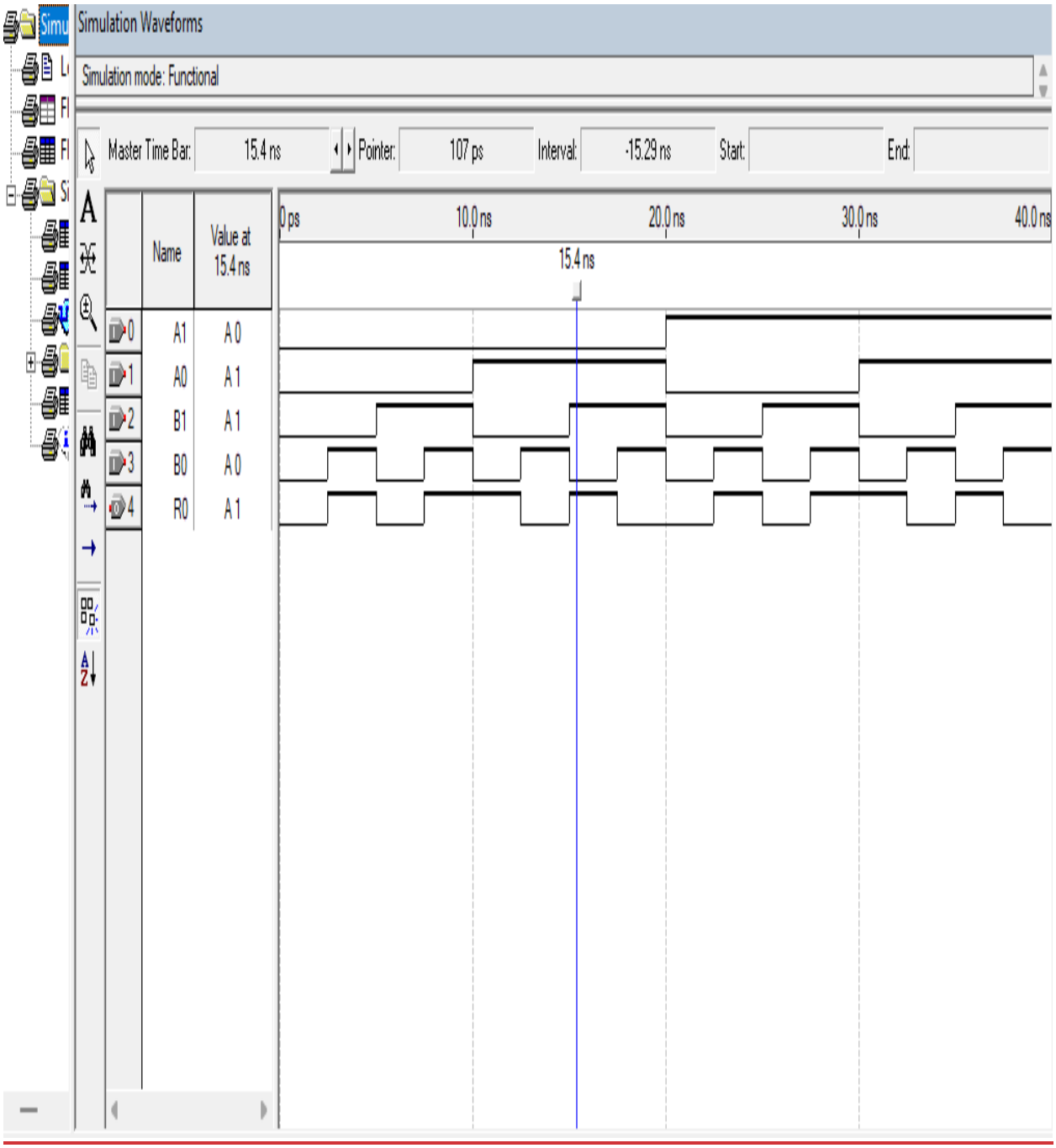


Figure 10: Simulation output

Discussion on Workflow:

The implementation of the 2-bit full adder using multiplexers was carried out in a systematic manner. First, the complete truth table for the 2-bit addition was constructed to understand the relationship between inputs and outputs. Based on this, separate truth tables were derived for Sum1, Sum2, and Carry.

Next, the Boolean functions for each output were analyzed and mapped into multiplexer form. The 8:1 multiplexer was used to implement the Sum1 function by assigning appropriate values to its inputs according to the truth table. Similarly, 2:1 multiplexers were used to generate the Sum2 outputs and Carry function by selecting between different logic conditions.

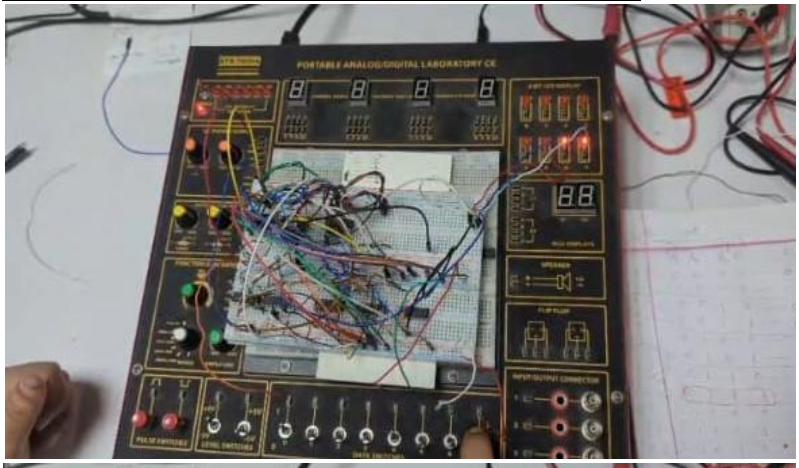
After designing the logic, the circuit was implemented using the cascade method, where the carry generated from the first stage was propagated to the second stage. Proper selection lines and input configurations were carefully arranged to ensure correct functionality.

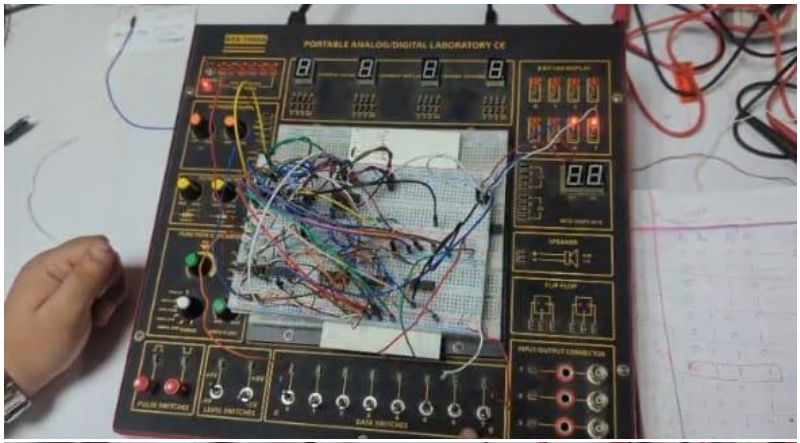
The circuit was then tested through simulation, where different input combinations were applied and the outputs were verified with the expected results from the truth table. The results matched successfully, confirming the correctness of the design.

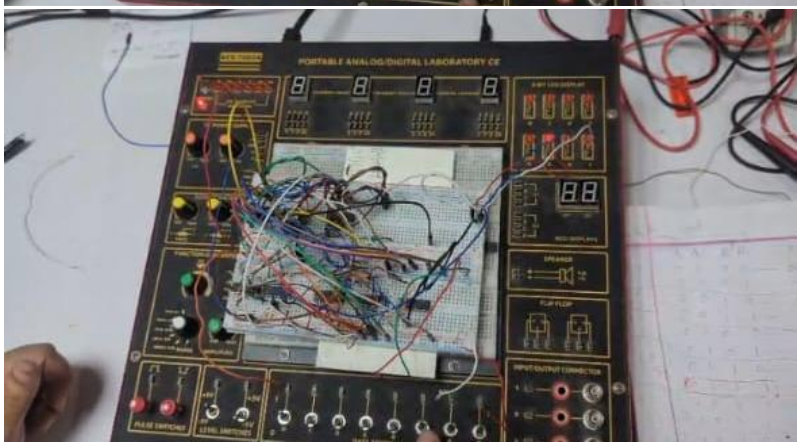
During the process, some challenges were faced, such as correctly mapping the truth table values into multiplexer inputs and managing carry propagation between stages. These were resolved through careful analysis and verification.

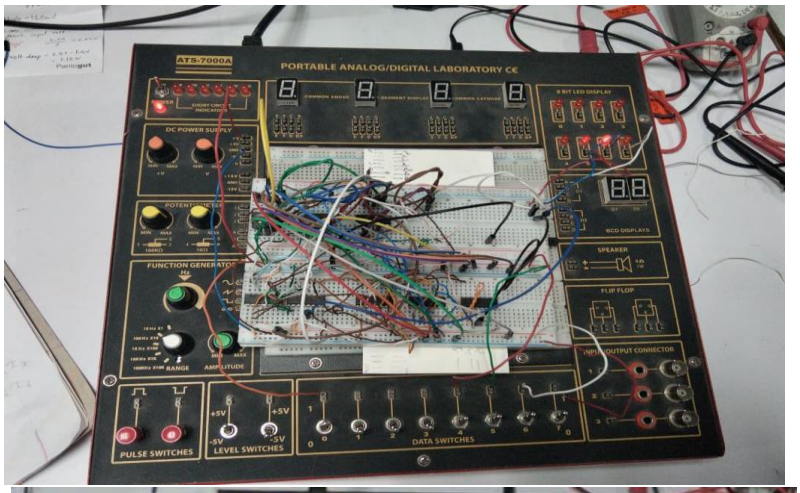
Overall, the workflow demonstrates a structured approach to designing combinational circuits using multiplexers, highlighting both theoretical understanding and practical implementation.

Observation of different states:

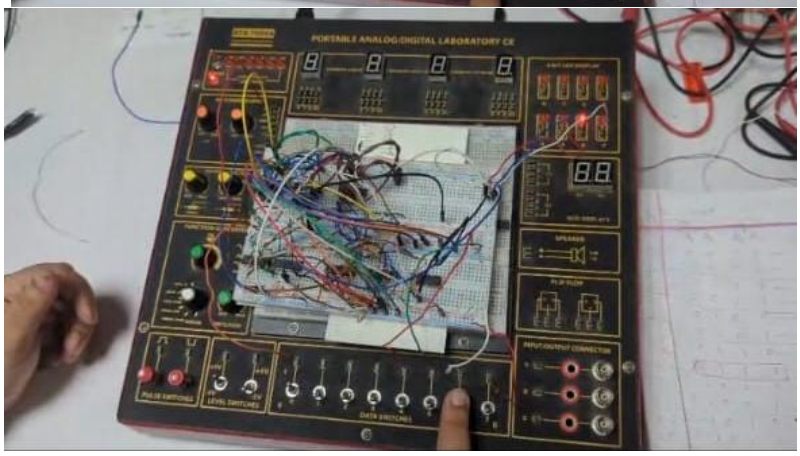







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LEVEL SWITCHES
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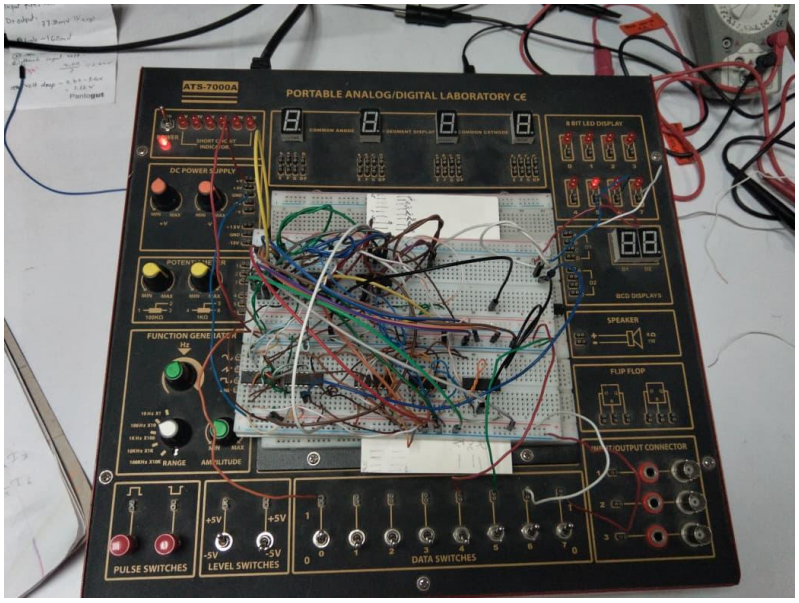


Figure: 6 diifferent states

Conclusion:

In this project, a 2-bit full adder was successfully designed and implemented using multiplexers with the cascade method. Instead of using conventional logic gates, the required sum and carry outputs were generated using an 8:1 multiplexer and 2:1 multiplexers.

The design demonstrated that multiplexers can effectively implement complex combinational logic circuits by properly mapping truth table values. The cascade approach ensured correct carry propagation between stages without the need for an external carry input.

The simulation results matched the theoretical outputs, confirming the accuracy of the design. Overall, this project provided a clear understanding of how multiplexers can be used as an alternative to logic gates and highlighted their importance in digital system design.

Reference:

1. https://web2.qatar.cmu.edu/cs/15348/labs/02/lab02.pdf
2. https://www.altera.com/downloads/fpga-development-tools/quartus-ii-web-edition-design-software-version-13-1-windows
3. https://www.youtube.com/watch?v=n55jedo8tRc
4. https://www.youtube.com/watch?v=AOQZgtWVZq0